Project Design Phase-II
Technology Stack (Architecture & Stack)

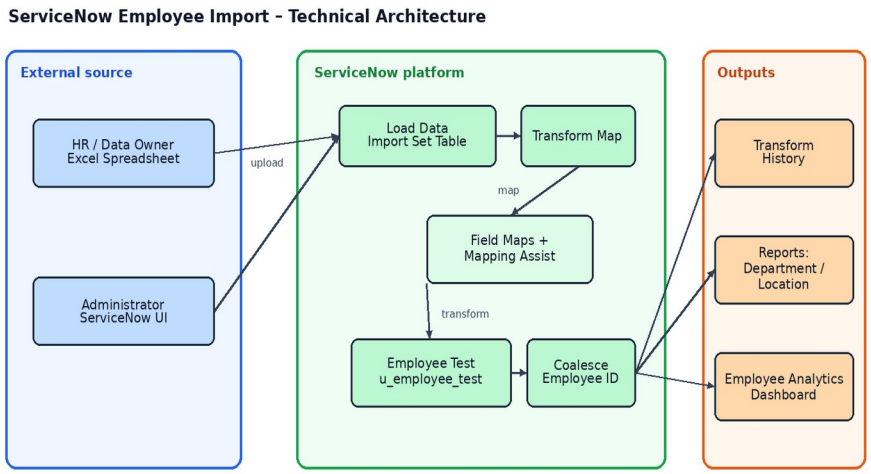
Date	30 September 2026
Team ID	SN-TRANSFORM-01
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows how the spreadsheet, ServiceNow staging and target tables, Transform Map logic, history, reports and dashboard work together.

ServiceNow Employee Import Architecture

Project reference: Import Data using Transform Maps (Spreadsheet)



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	Source Interface	Spreadsheet supplied by HR / data owner; header row and sample employee records.	Excel / Google Sheets export
2	Import Set Table	Staging area that receives the uploaded rows before transformation.	ServiceNow Import Set: Employee Import
3	Transform Map	Controls source-to-target mapping and executes the employee data transformation.	Sample Spreadsheet Import
4	Field Mapping	Maps Employee ID, Name, Email, Department and Location between tables.	Auto Map + Mapping Assist
5	Target Database	Stores validated employee records in the custom target table.	ServiceNow: Employee Test / u_employee_test
6	Duplicate Control	Matches existing records so repeated IDs update rather than insert duplicates.	Coalesce = true on Employee ID
7	Audit / History	Records transform state and inserted, updated, ignored and error outcomes.	Transform History
8	Reporting	Visualizes department distribution, location distribution and employee records.	ServiceNow Reports: Pie, Bar, List
9	Dashboard	Combines the three reports for HR and administrator monitoring.	Employee Analytics Dashboard
10	Access Control	Controls who can load data, configure maps, view records and manage reports.	ServiceNow roles and ACLs
11	Infrastructure	Managed application and database services host the import workflow.	ServiceNow platform / SaaS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-source frameworks	No custom application framework is required; the deliverable uses standard ServiceNow configuration.	ServiceNow platform configuration
2	Security implementations	Use role-based access, ACLs, controlled imports and Employee ID as the logical match key.	ServiceNow ACLs, roles, Coalesce
3	Scalable architecture	Staging separates file intake from target storage and supports larger recurring imports.	Import Sets + Transform Maps
4	Availability	Reports and dashboard are centrally available to authorized ServiceNow users.	ServiceNow application services
5	Performance	Batch load and transform reduce manual entry; mapping and validation keep data processing focused.	Import Set batch processing

References:

ServiceNow project brief: Import Data using Transform Maps (Spreadsheet)

ServiceNow Import Sets and Transform Maps configuration workflow

ServiceNow reporting and dashboard configuration

Custom table u_employee_test and Import Set u_employee_import

Design principle: stage, transform, validate, report